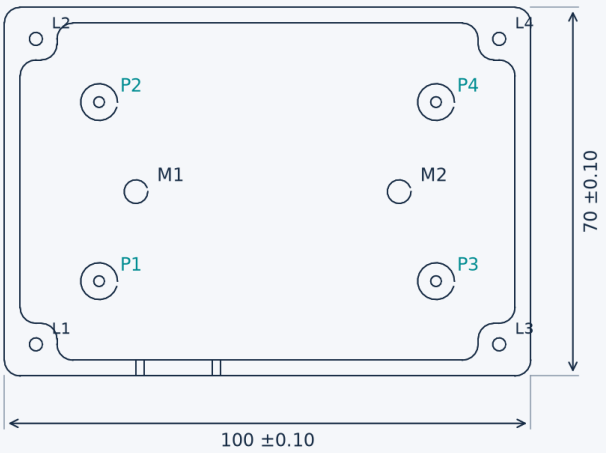
DRAWING PACKAGE / REV C / REVIEW ISSUE / mm

E01 / Base enclosure

E01 / Base enclosure | mm



A: underside Z0 | B: front Y0 | C: left X0

Hole coordinates and tolerances in table below.

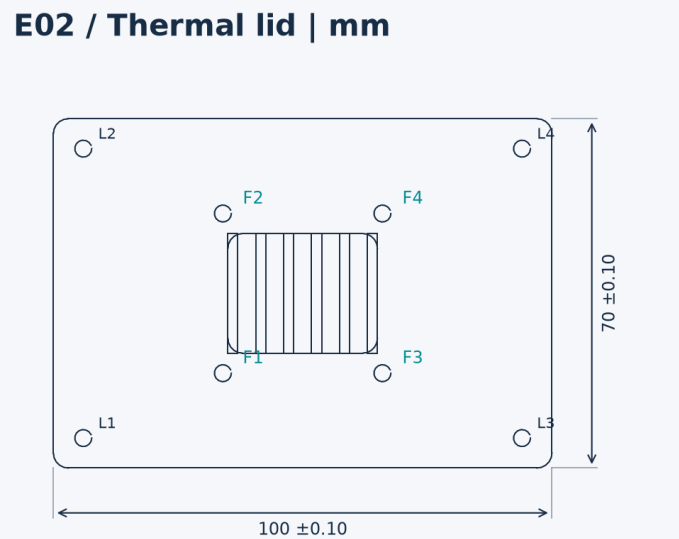
Top projection; underside and pocket edges included.

Feature / nominal	Location / requirement
Envelope 100 × 70 × 17	X/Y±0.10; height ±0.03 from datum A; exterior R3
Pocket / floor	Wall 3.00 min nominal; floor 3.00±0.10; cavity R3;10 mm corner islands
L1..L4 / M3 × 0.5	(6,6),(6,64),(94,6),(94,64); position Ø0.10 to A/B/C; target full engagement 6
P1..P4 / M2.5 × 0.45	(18,18),(18,52),(82,18),(82,52); support top Z6.00±0.03; position Ø0.10
M1..M2 / Ø4.50 +0.10/-0	(25,35),(75,35); through floor; position Ø0.20 to A/B/C
Front opening	X25..41, Z7..14, R1.50; size+0.15/-0, location±0.10

Datums: A underside Z0; B front Y0; C left X0. Proposed A flatness 0.05; support coplanarity 0.05. CNC 6061-T6; edge break 0.2–0.4 not modeled. Thread pilots omit real drill-point geometry: shop must verify blind depth and remaining floor.

CNC 6061-T6 / ALL DIMENSIONS mm

E02 / Integral thermal lid

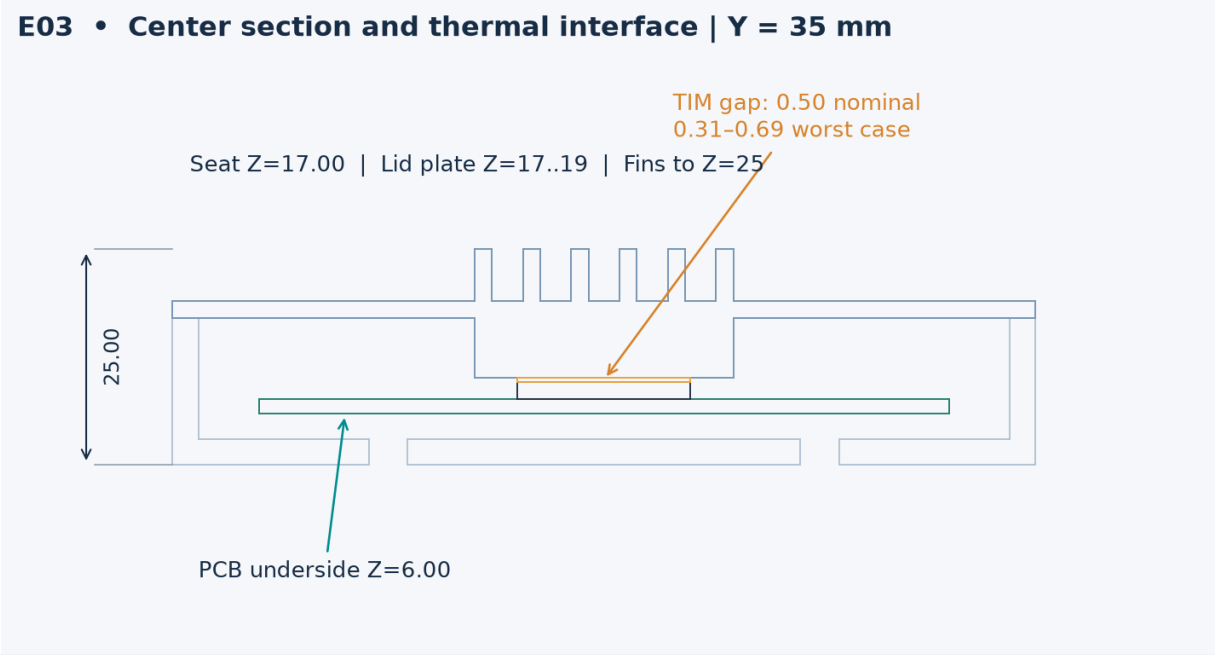


Feature	Dimension / control
Outer profile	100 × 70±0.10; vertical corners R3
Mating plane D	Assembly Z17; flatness 0.05; front E / left F locate X/Y
Plate thickness	2.00±0.05 above D; top Z19
Pedestal underside	30 × 24, R3, centered (50,35);6.90±0.03 below D; contact flatness 0.03
Fin field	X35..65, Y23..47; six 2.00±0.10 fins,6.00±0.10 high;3.60 channels
Four lid holes	Ø3.40 +0.10/-0 at (6,6),(6,64),(94,6),(94,64); position Ø0.15 to D/E/F
Four fan holes	Ø3.40 +0.10/-0 at (34,19),(34,51),(66,19),(66,51); position Ø0.20 to D/E/F

Flatness controls seating/contact; position controls simultaneous screw assembly. Inspect with calibrated height/plane tools and qualified hole-center measurement. Support the 2 mm plate while machining. No specified surface-roughness result is claimed; supplier pad requirements must set the contact finish before release.

NOMINAL ASSEMBLY / NOT A PHYSICAL MEASUREMENT

E03 / Assembly section & fits

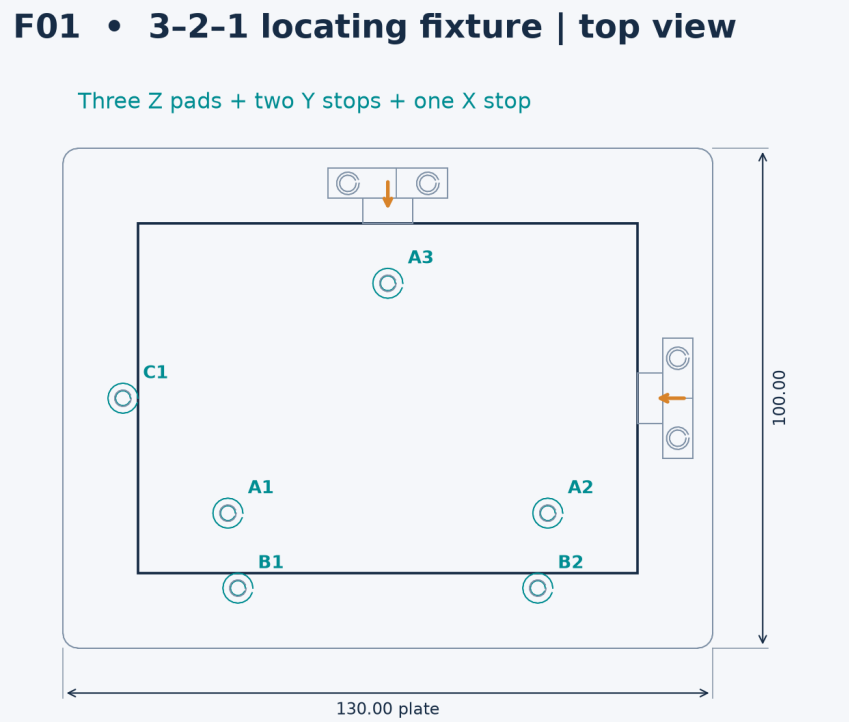


Interface	Nominal / acceptance proposal
Board	80 × 50 × 1.6; lower Z6; holes Ø2.7 assumed; chip top Z9.6
TIM	Compressed gap 0.50; worst case 0.31..0.69; supplier force curve unresolved
Secondary components	Top Z12.6 vs underside lid Z17 =>4.4 nominal clearance outside pedestal
Connector	0.75 nominal XZ bounding-box margin; R1.5 opening corner checked in B-rep
External mount	M4 head height ≤2.5 above floor Z3; reserve underside-board keepouts
Fasteners	Lid M3 × 8; PCB M2.5 × 5; simplified heads/roots; threads not explicitly modeled
Fan option	40 × 40 × 10 frame at Z25..35; four 6 mm spacers; reserve hardware to Z38

Stack g =17.00-6.90-6.00-1.60-2.00. Worst case±0.19. M3 lid clearance radial allowance 0.20; nominal base/lid position radii consume 0.125. Printed fits require independent calibration and are not subject to CNC acceptance tolerances.

3-2-1 / GRAVITY-SEATED BENCH INSPECTION

F01 / Inspection fixture



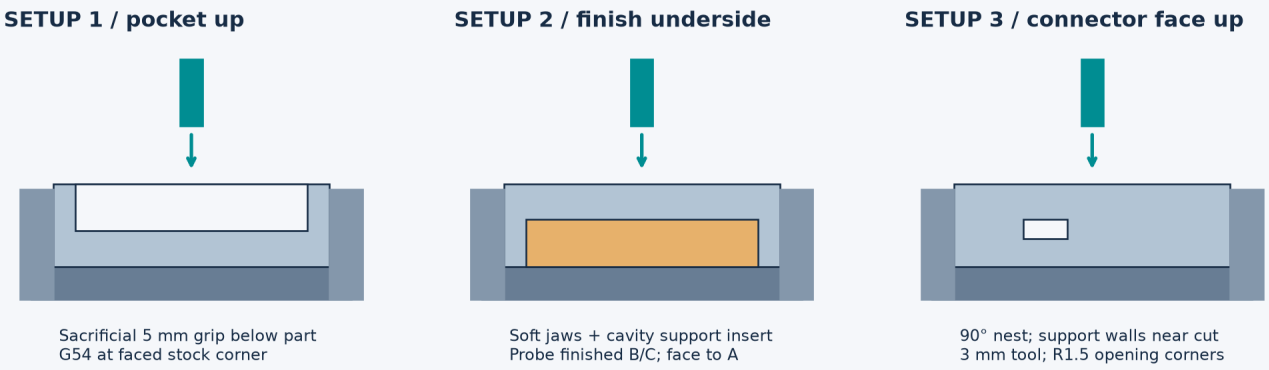
Element	Coordinates / geometry
Plate	130 × 100 × 6; origin corner (-15,-15,-10); top Z-4
Three support buttons	(18,12),(82,12),(50,58);Ø6; top Z0
Two B / one C stops	(20,-3),(80,-3),(-3,35);Ø6; top Z5
Mounting stems	Simplified Ø3 to Z-14 through Ø3.3 plate holes; underside nuts/washers required
Clamp block mounts	M4 plate pilots at (42,78),(58,78),(108,27),(108,43); blocks Ø4.5 clearance

Locator coordinates±0.10 initial fabrication target; calibrate with a reference tray. Shim A buttons to≤0.02 height range before repeatability study. Horizontal M4 screw pilots drive replaceable pads; screws/feet/hold-down not fully modeled. Do not use as a machining fixture.

PLANNING ARTIFACT / NO VERIFIED TOOLPATHS

M01 / CNC setup reference

CNC setup concepts | review before programming



Illustrative setup geometry. No generated or verified machine toolpaths; clamp and tool envelopes require CAM review.

Operation	Review before cutting
Top pocket	Sacrificial grip 5 mm;6 mm rougher,14 mm cavity depth; maintain islands
Flip / underside	Cavity support nest; confirm B/C transfer; measure A flatness after unclamping
Front opening	90-degree supported nest;3 mm cutter through wall; inspect burrs/corner clearance
Two-sided lid	Relieved support nest; clamp outside finished thin plate; review fin tool reach
Machine prove-out	Verify exact machine/post, tools, offsets, clamps, collisions, retracts and first-article inspection

These diagrams explain workholding intent; they are not to scale and do not establish safe clamping forces. The complete operation sequence and checklist are in manufacturing/cnc/setup-plan.md. A qualified shop must generate and simulate the actual CAM setup.